

A formal proof of the E-L* noisy DFA learner

(pseudocode level, all assumptions explicit)

Abstract

This is the rigorous companion to the informal correctness note. It states the model, the algorithm, and the proof at a level where every step is either proved or reduced to a numbered assumption. The intent is to force out the sub-facts the informal convergence argument leaned on without naming. Where a step is genuinely an assumption about the target or the sampler and not a theorem, it is boxed as one; Section 9 collects them and marks which are load-bearing. Two proofs need external theorems: the exact certification test cites Hoeffding (1956); everything else is self-contained.

1 Model

1.1 Target and oracle

The target is a language $L \subseteq \Sigma^*$ recognised by the minimal DFA $A^* = (Q^*, \Sigma, t, q_0, F)$. Write $q^*(u)$ for the state u reaches and x_q for a fixed access string with $q^*(x_q) = q$. The Myhill–Nerode property is that membership of any extension of u depends on u only through $q^*(u)$:

$$q^*(u) = q^*(u') \implies (uw \in L \iff u'w \in L) \quad \forall w. \quad (1)$$

Definition 1 (Persistent signal oracle). *Fix a base rate $\beta \in (0, 1)$ and signal $s \in (0, \min(\beta, 1 - \beta)]$. For each string x the oracle holds a fixed bit $\text{MQ}(x) \in \{0, 1\}$, drawn once and independently across distinct strings, with*

$$\Pr[\text{MQ}(x) = 1] = \begin{cases} \beta + s & x \in L, \\ \beta - s & x \notin L. \end{cases}$$

The bit is persistent: re-querying x returns the same value, so a single string yields one read.

The independence-across-distinct-strings clause is the whole reason a vote denoises; it is a property of the model, used explicitly below.

Definition 2 (Access and goal). *The learner may call MQ on any string and may sample from a distribution D over Σ^* . It outputs a DFA A ; the target accuracy is $\Pr_{x \sim D}[A(x) = A^*(x)] \geq 1 - \varepsilon$ with probability $\geq 1 - \delta$ over the oracle draw and the learner’s randomness.*

1.2 Midfixes, cuts, families

Definition 3 (Cut). *For a midfix $m \in \Sigma^*$, its cut is $D_m = \{q \in Q^* : x_q m \in L\}$; by (1) this is independent of the choice of access string.*

A node of the discrimination tree carries a midfix m and routes u left if $q^*(u) \in D_m$, else right; leaves are classes of states the tree has not yet separated. The learner cannot evaluate $q \in D_m$ directly — one read $\text{MQ}(x_q m)$ carries signal only s — so it votes a family of suffixes.

Definition 4 (Accept-preserving). A suffix v is accept-preserving on a set $R \subseteq Q^*$ if $x_q v \in L \iff x_q \in R$ for every $q \in R$. The empty suffix ε is accept-preserving on all of Q^* .

Definition 5 (Vote and decision). For a family $V = (v_1, \dots, v_k)$ of distinct suffixes, split β and margin $\tau \in (0, s)$, the vote and decision of a string x are

$$\bar{r}(x) = \frac{1}{k} \sum_{i=1}^k \text{MQ}(x v_i), \quad \text{decide}(x) = \begin{cases} \text{ACCEPT} & \bar{r}(x) > \beta + \tau, \\ \text{REJECT} & \bar{r}(x) < \beta - \tau, \\ \text{INDECISIVE} & \text{otherwise.} \end{cases}$$

A node with midfix m classifies u by $\text{decide}(u m)$.

2 Denoising: soundness of a decisive decision

Lemma 1 (Independence of a vote). For distinct $v_1, \dots, v_k$ the strings $xv_1, \dots, xv_k$ are distinct, so the reads $\text{MQ}(xv_i)$ are independent (Def. 1). Hence $k \bar{r}(x) = \sum_i \text{MQ}(xv_i)$ is a sum of k independent Bernoulli variables with means $\mu_i = \beta + s \cdot (2 \cdot \mathbf{1}[xv_i \in L] - 1)$.

Proof. Distinctness of the v_i gives distinct query strings; Definition 1 makes the bits independent, and each has the stated mean by membership of xv_i . $\square$

Definition 6 (Error rates). Put $\mu(x) = \mathbb{E}[\bar{r}(x)] = \frac{1}{k} \sum_i \mu_i$. Define

$$\text{fpr}(k, \tau) = 2e^{-2k\tau^2}, \quad \text{fnr}(k, \tau) = e^{-2k(s-\tau)^2}.$$

Lemma 2 (Denoising bounds). For any string x and any family:

- (i) if $\mu(x) \leq \beta$ then $\Pr[\text{decide}(x) = \text{ACCEPT}] \leq e^{-2k\tau^2}$, and symmetrically for REJECT when $\mu(x) \geq \beta$; so a decision on the wrong side of β has probability $\leq \text{fpr}(k, \tau)$;
- (ii) if x is such that all xv_i share membership b (the family is accept-preserving for x), then $\mu(x) = \beta \pm s$ and $\Pr[\text{decide}(x) = \text{INDECISIVE}] \leq \text{fnr}(k, \tau)$.

Proof. $k\bar{r}(x)$ is a sum of independent $[0, 1]$ variables (Lemma 1) with mean $k\mu(x)$. (i) $\{\text{decide} = \text{ACCEPT}\} = \{\bar{r} > \beta + \tau\}$ needs $\bar{r} - \mu(x) > \beta + \tau - \mu(x) \geq \tau$; Hoeffding's inequality gives $e^{-2k\tau^2}$. (ii) With common membership b , $\mu(x) = \beta + s$ (say); $\{\text{INDECISIVE}\} \subseteq \{\bar{r} \leq \beta + \tau\}$ needs $\bar{r} - \mu(x) \leq -(s - \tau)$, of probability $\leq e^{-2k(s-\tau)^2}$. $\square$

Lemma 3 (A decisive decision realises the noiseless rule). For any V, β the map

$$\text{decide}^*(q) = \begin{cases} \text{ACCEPT} & \mu^*(q) > \beta, \\ \text{REJECT} & \mu^*(q) < \beta, \end{cases} \quad \mu^*(q) = \mathbb{E}[\bar{r}(x_q m)],$$

is well defined and depends on q alone. For every u , $\Pr[\text{decide}(u m) \neq \text{decide}^*(q^*(u)) \text{ and } \text{decide}(u m) \neq \text{INDECISIVE}] \leq \text{fpr}(k, \tau)$.

Proof. $\mu^*(q) = \frac{1}{k} \sum_i \mathbb{E}[\text{MQ}(x_q m v_i)]$, and $\mathbb{E}[\text{MQ}(x_q m v_i)]$ is $\beta + s$ or $\beta - s$ according to $x_q m v_i \in L$, which by (1) depends only on $q^*(x_q) = q$; so μ^* is a function of q . A decisive disagreement — say ACCEPT while $\mu^*(q^*(u)) < \beta$ — is the event of Lemma 2(i) with $x = u m$, of probability $\leq \text{fpr}$. $\square$

Soundness holds for any family: whether or not V is accept-preserving, a decision it makes decisively agrees with a genuine state predicate except with probability fpr . The family only controls coverage — the fraction of strings decided rather than left INDECISIVE (Lemma 2(ii) needs the family accept-preserving for x). Coverage is where correctness could silently fail, and is what the populations and the certificate below govern.

3 Prefix populations

The learner reads its rates not over one sample but over a finite collection $\mathcal{P} = \{P_1, \dots, P_r\}$ of populations of prefixes, held to each rate separately:

- the uniform pool P_{unif} , drawn i.i.d. from D ;
- for each leaf q of the current tree, P_q : prefixes the tree rests at q (the tree's own placement, not the hypothesis' aim);
- for each past round, a boundary population of strings that round left INDECISIVE.

A prefix may lie in several. The reason to separate them is arithmetic and is the crux of the whole design:

Lemma 4 (A union rate hides a minority). *Let a family read a fraction ρ_i of population P_i as INDECISIVE (or, for the certificate below, as the wrong class). The pooled rate over $\bigcup_i P_i$ is $\sum_i \rho_i |P_i| / \sum_i |P_i|$. If P_j is entirely misread ($\rho_j = 1$) while all others are clean, the pooled rate is $|P_j| / \sum_i |P_i|$, which is below any limit ℓ whenever $|P_j| < \ell \sum_i |P_i|$. Thus no bound on the pooled rate constrains a population smaller than an ℓ share of the union.*

Proof. Immediate from the definition of the pooled average. $\square$

Every rate below — clustering loss, FNR, certification — is therefore stated per population and enforced as a maximum, $\max_i(\cdot)$, not an average. A population the learner cannot fill (a leaf no sampled string reaches) is dropped; Assumption 2 says which populations can be filled.

4 The algorithm

Algorithm 1 CLUSTER(β): ε -anchored family and refreshed split

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1: fill  $M[v][j] = \text{MQ}(x_j v)$  over candidate suffixes  $v$  and pooled prefixes  $j$ 
2:  $w_j \leftarrow 1/|P(j)|$   $\triangleright P(j)$  = the population of prefix  $j$ ; equal say per population
3:  $V \leftarrow \{\varepsilon\}$ ;  $\ell^* \leftarrow \infty$ 
4: loop
5:    $c[j] \leftarrow \mathbf{1}[\text{mean}_{v \in V} M[v][j] > \beta]$ 
6:    $N \leftarrow$  the  $k$  rows minimising  $\ell(v) = \sum_j w_j \mathbf{1}[M[v][j] \neq c[j]]$ 
7:   if  $\varepsilon \notin N$  or  $\sum_{v \in N} \ell(v) \geq \ell^*$  then break
8:    $V \leftarrow N$ ;  $\ell^* \leftarrow \sum_{v \in N} \ell(v)$ 
9:  $\beta \leftarrow \frac{1}{2}(\bar{r}_{\text{acc}} + \bar{r}_{\text{rej}})$   $\triangleright$  midpoint of the mean vote on  $V$ 's two sides
10: return  $(V, \beta)$ 
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5 Certification

Fix one side of the cut of a candidate family, restricted to one population: n prefixes $x_1, \dots, x_n$ land there, and $H = \sum_j M[\varepsilon][j]$ of them read 1 on the empty-suffix column. Since $M[\varepsilon][j] = \text{MQ}(x_j)$,

Algorithm 2 SAMPLEFAMILY: resampling loop with per-population gates

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1: loop
2:    $(V, \beta) \leftarrow \text{CLUSTER}(\beta); \quad \tau \leftarrow \text{margin meeting fpr, fnr budgets at } (\beta, k)$ 
3:    $\widehat{\text{fnr}}_P \leftarrow \text{fraction of } P \text{ left INDECISIVE by } (V, \beta, \tau), \text{ each population } P$ 
4:    $P^* \leftarrow \arg \max_P \widehat{\text{fnr}}_P$ 
5:   if  $|V| < k$  or  $\widehat{\text{fnr}}_{P^*} > \delta_{\text{fnr}}$  then
6:     draw more suffixes (screened against  $M[\varepsilon]$ ) or grow  $P^*$ ; continue
7:   on the column  $M[\varepsilon]$ , count each population's two sides of the cut
8:   if some population is certified read as the opposite class then refuse; continue ▷ veto
9:   else if  $P_{\text{unif}}$ 's two sides are certified each on its own class then
10:    return  $(V, \beta, \tau)$  ▷ admit
11:   else draw more of  $P_{\text{unif}}$  for the split; continue ▷ undecided

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Algorithm 3 LEARN: the round loop

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1: populations  $\leftarrow \{P_{\text{unif}}\}$  from an i.i.d.  $D$ -sample; tree  $\leftarrow$  single leaf
2: loop
3:    $(V, \beta, \tau) \leftarrow \text{SAMPLEFAMILY}()$  anchored at  $\varepsilon$ 
4:   build hypothesis  $(A, \text{tree})$  by resolving transitions with  $(V, \beta, \tau)$ 
5:   if  $A$  is consistent enough, or the round harvested no new boundary strings then return  $A$ 
6:   rebuild populations: refresh  $P_{\text{unif}}$ , harvest boundary strings, sample each leaf

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the reads are independent (Def. 1) with $\mathbb{E}[M[\varepsilon][j]] = \beta + s$ if $x_j \in L$ and $\beta - s$ if not. If a fraction f of the n are really the other class, the mean read is

$$\bar{p} = (1 - f)(\beta + s) + f(\beta - s) = \beta + s - 2sf. \quad (2)$$

The tests are $\Pr[\text{Bin}(n, \beta + \tau) \geq H] \leq \alpha$ (accept side reads high) and $\Pr[\text{Bin}(n, \beta - \tau) \leq H] \leq \alpha$ (reject side reads low). The difficulty is that H is a sum of non-identically-distributed Bernoullis, so a scalar monotonicity argument does not apply. We isolate the fact that does.

Lemma 5 (MGF domination). *Let $X_1, \dots, X_n$ be independent, $X_j \sim \text{Bin}(1, p_j)$, $\bar{p} = \frac{1}{n} \sum_j p_j$, $H = \sum_j X_j$. For every $\lambda > 0$,*

$$\mathbb{E}[e^{\lambda H}] = \prod_j (1 + p_j(e^\lambda - 1)) \leq (1 + \bar{p}(e^\lambda - 1))^n = \mathbb{E}[e^{\lambda B}], \quad B \sim \text{Bin}(n, \bar{p}).$$

Consequently, for any $q \geq \bar{p}$ and any threshold h ,

$$\Pr[H \geq h] \leq \inf_{\lambda > 0} e^{-\lambda h} (1 + q(e^\lambda - 1))^n = \exp(-n \text{KL}(h/n \| q)) \quad (h/n \geq q),$$

the Chernoff upper-tail bound of $\text{Bin}(n, q)$.

Proof. Independence gives the product form. Each factor $a_j = 1 + p_j(e^\lambda - 1) > 0$, and by AM–GM $\prod_j a_j \leq (\frac{1}{n} \sum_j a_j)^n = (1 + \bar{p}(e^\lambda - 1))^n$. The map $q \mapsto 1 + q(e^\lambda - 1)$ is increasing for $\lambda > 0$, so raising $\bar{p}$ to q only enlarges the bound; optimising λ gives the stated Chernoff/KL form. $\square$

Proposition 1 (Certification is sound: Chernoff form). *Suppose a side is admitted only when $H/n > \beta + \tau$ and the $\text{Bin}(n, \beta + \tau)$ Chernoff upper-tail bound at H is $\leq \alpha$, i.e. $\exp(-n \text{KL}(H/n \| \beta + \tau)) \leq \alpha$.*

$\tau)) \leq \alpha$. If the side is drifted at or past the bar, $f \geq \frac{s-\tau}{2s}$ (equivalently $\bar{p} \leq \beta + \tau$ by (2)), then $\Pr[\text{admit}] \leq \alpha$. Symmetrically for the reject side and the veto tests. Hence, on admission, with confidence $\geq 1 - \alpha$ the side carries

$$f < \frac{s - \tau}{2s}$$

of the other class.

Proof. On admission $H/n > \beta + \tau \geq \bar{p}$, so Lemma 5 with $q = \beta + \tau$ gives $\Pr[H \geq H_{\text{obs}}]$ bounded by the $\text{Bin}(n, \beta + \tau)$ Chernoff bound, which admission forces $\leq \alpha$. The contrapositive is the confidence statement, and $\bar{p} \leq \beta + \tau \iff f \geq (s - \tau)/2s$ by (2). $\square$

Remark 1 (The exact binomial test). *The implementation uses the exact tail $\Pr[\text{Bin}(n, \beta + \tau) \geq H]$ rather than its Chernoff bound. Validity of the exact test needs $\Pr[H \geq h] \leq \Pr[\text{Bin}(n, \bar{p}) \geq h]$ for the non-i.i.d. H , which is Hoeffding’s 1956 tail theorem, valid for $h \geq n\bar{p} + 1$. At the admission threshold $h > n(\beta + \tau) \geq n\bar{p}$, and for the levels α in use the threshold clears $n\bar{p} + 1$, so the exact test is valid there; combined with binomial monotonicity in the rate ($\bar{p} \leq \beta + \tau$) it gives the same α guarantee. The self-contained core (Prop. 1) does not depend on this refinement.*

Proposition 2 (Only the base-rate pool may admit). *Admission is the claim that the family separates the two classes: both sides read as their own class. This is testable only on a population that contains both classes in known proportion. A per-state population P_q is a single class ($f \in \{0, 1\}$), so one of its sides is empty and its admission test is vacuous; it can veto (its one side read against its class) but cannot attest to a separation it never straddles. The uniform pool is the population whose class proportions equal the target’s base rate under D ; admission is read only there. Veto, needing only that a populated side reads against its class, is read on every population.*

Proof. A side with $n = 0$ triggers neither one-sided test, so a single-class population cannot satisfy the two-sided admission conjunction; it satisfies a veto test as in Prop. 1 applied to its populated side. That the uniform pool’s proportions match D is by construction (it is an i.i.d. D -sample), so its two-sided reading estimates the separation the goal in Def. 2 is scored against. $\square$

6 The sampling loop halts and returns a certified family

Assumption 1 (Findable accept-preserving suffix). *Each suffix draw is accept-preserving on Q^* with probability $\geq p_{\text{preserve}} > 0$.*

Assumption 2 (Coverage). *There is a covered set $R \subseteq Q^*$ with (i) $\Pr_{x \sim D}[q^*(x) \notin R] \leq \varepsilon_{\text{cov}}$, and (ii) every $q \in R$ is reached by a D -draw with probability $\geq p_{\text{reach}} > 0$. L is nontrivial: both classes have positive D -mass, so the uniform pool contains members and non-members.*

Assumption 3 (Screening separability). *For every suffix v not accept-preserving on R there is $q \in R$ with $\mathbf{1}[x_q \in L] \neq \mathbf{1}[x_q v \in L]$; and the screen that compares a candidate v against $M[\varepsilon]$ over the populations rejects v once a prefix reaching q is present in some population — which occurs with probability $\geq p_{\text{reach}}$ per draw by Assumption 2(ii).*

Theorem 1 (SAMPLEFAMILY returns a certified family). *Under Assumptions 1–3 the loop of Algorithm 2 halts with probability 1, and the family it returns (a) leaves at most δ_{fmr} of every population INDECISIVE, (b) is admitted by the uniform pool with no population vetoing, so by Proposition 1 each side of its cut carries $f < (s - \tau)/2s$ of the other class with confidence $\geq 1 - \alpha$. When no suffix is accept-preserving on R , the uniform pool never admits and the idealised loop does not halt.*

Proof. Halting. Consider the events across draws. By Assumption 1 an accept-preserving suffix is drawn infinitely often almost surely (Borel–Cantelli, independent draws each $\geq p_{\text{preserve}}$). By Assumption 3, any drawn suffix flipping a covered state is rejected once that state is populated, and each covered state is populated after finitely many draws a.s. (Assumption 2(ii), Borel–Cantelli). Hence after finitely many draws a.s. the candidate pool contains k suffixes accept-preserving on R ; the ε -anchored cluster (Alg. 1), which retains ε and whose weighted loss (Lemma 4, weighting) gives each covered state equal say, then proposes a family driftless on R . For such a family every covered prefix is coherent, so by Lemma 2(ii) taking k large enough (finite, by Def. 6) drives $\text{fmr}_P \leq \delta_{\text{fmr}}$ on each fillable P ; a population still short is grown (P^* lever), which succeeds by Assumption 2(ii). A driftless family has no side read against its class, so no population vetoes (Prop. 1); and its uniform pool, containing both classes (Assumption 2), has both sides on their own class, so after finitely many split reads the admission conjunction holds (Prop. 2). The loop returns.

Guarantees. The loop returns only on the admit branch, which requires (a) and (b) verbatim; the confidence bound is Proposition 1.

No accept-preserving family. If every candidate flips some covered state on positive mass, the uniform pool’s separation is contradicted on that mass, so the two-sided admission test fails for every proposal; the idealised loop never takes the admit branch. (The implementation caps the loop with a give-up budget spent only on proposals a population has vetoed, then reports the target has none.) $\square$

7 Rounds converge to the covered state partition

Assumption 4 (Distinguisher findability). *Whenever two covered states $q \neq q' \in R$ share a leaf of the current tree, each round draws, with probability $\geq p_{\text{dist}} > 0$, prefixes reaching q and q' and a midfix m with exactly one of q, q' in D_m (Def. 3).*

Assumption 5 (Resolver specification). *Given a certified family (Thm. 1) and a midfix m that is decisive (Def. 5) and separates two states currently sharing a leaf, the resolver splits that leaf into the two sides of m ’s cut and never merges leaves; the hypothesis A it emits routes each covered state to its leaf consistently with the tree.*

Lemma 6 (Progress). *Under Assumptions 2–5, while some leaf holds two covered states, some round a.s. splits a leaf.*

Proof. Fix such a pair q, q' . By Assumption 4 a round exposes a separating midfix m with prefixes reaching each, and those prefixes form (or join) the per-state populations $P_q, P_{q'}$. The returned family is decisive on every population to δ_{fmr} (Thm. 1(a)), so all but a δ_{fmr} fraction of $P_q \cup P_{q'}$ are decided; a decided string at m is $\text{decide}(u m)$, which by Lemma 3 equals $\text{decide}^*(q^*(u))$ except w.p. fpr , and since exactly one of q, q' lies in D_m the two states fall on opposite decisive sides. So a decisive separating m is present; by Assumption 5 the resolver splits the leaf. Assumption 4 recurs each round, so by Borel–Cantelli a split occurs a.s. $\square$

Theorem 2 (Convergence to the covered partition). *Under Assumptions 1–5, with probability 1 after finitely many rounds the leaves of the tree are exactly the covered states: each $q \in R$ occupies its own leaf, and two states in one leaf are Myhill–Nerode-equal.*

Proof. Splits refine and never merge (Assumption 5), so the number of leaves is non-decreasing and bounded by $|Q^*|$. By Lemma 6 each round with a covered pair sharing a leaf a.s. eventually splits one; a split strictly increases the leaf count, so after at most $|R| - 1$ effective splits no covered pair

shares a leaf. Two states the tree separates are separated by a cut D_m containing exactly one of them, hence Myhill–Nerode-distinct (Def. 3, (1)); so distinct covered leaves are distinct states, and each covered leaf is a singleton. $\square$

8 Main theorem

Theorem 3 (Correctness of LEARN). *Assume 1–5. Let d bound the tree depth (so $d \leq |Q^*|$). If k, τ are set so $\text{fpr}(k, \tau) \leq (\varepsilon - \varepsilon_{\text{cov}})/d$ and α and the round budget are set so the total failure probability over the finitely many certifications and rounds is $\leq \delta$, then LEARN (Alg. 3) outputs A with*

$$\Pr_{x \sim D}[A(x) \neq A^*(x)] \leq \varepsilon \quad \text{with probability } \geq 1 - \delta.$$

Proof. By Theorem 2 the tree’s leaves are the covered states after finitely many rounds a.s. Take $x \sim D$. If $q^*(x) \notin R$, it may be misclassified; this has D -mass $\leq \varepsilon_{\text{cov}}$ (Assumption 2(i)). If $q^*(x) \in R$, routing x traverses $\leq d$ decisive nodes, each returning x ’s true side except w.p. fpr (Lemma 3); by a union bound x reaches its true (singleton) leaf, whose state is $q^*(x)$, except w.p. $d \cdot \text{fpr}$. So

$$\Pr_{x \sim D}[A(x) \neq A^*(x)] \leq \varepsilon_{\text{cov}} + d \cdot \text{fpr}(k, \tau) \leq \varepsilon.$$

The oracle-level failures — a miscertified family (each $\leq \alpha$, Thm. 1), and non-convergence within the round budget — are made to sum to $\leq \delta$ by the choice of α and the budget, which is finite by Theorems 1 and 2. $\square$

9 Assumption ledger

The proof rests on exactly the following, in decreasing order of how much it carries and how far it is from provable:

- 2 Coverage. Load-bearing and irreducible: a state no D -draw reaches is never populated, never split, and can be misclassified; its mass appears as ε_{cov} in Theorem 3. This is the honest boundary of what any sampler-driven learner can promise.
- 4 Distinguisher findability. Load-bearing for convergence. It is the noiseless-separation signal: without it two distinct states can share a leaf forever. It is a statement about D and the midfix sampler, not about noise.
- 3 Screening separability. Load-bearing for halting and for which family is returned. This is the sub-fact the informal note waved at (“the screen drops any suffix flipping a state on positive D -mass”); made explicit, it is a consequence of Coverage plus the definition of the screen, so it is nearly derivable — it needs only that a covered flipped state is eventually populated, which is Assumption 2(ii).
- 1 Findable accept-preserving suffix. Load-bearing for halting; genuinely an assumption about D (some accept-preserving suffix has positive draw mass).
- 5 Resolver specification. Not a fact about the target but the interface contract of the transition resolver: it splits on a decisive separating midfix and never merges. Discharging it fully means verifying the resolver code; here it is the one place the pseudocode abstracts an implementation.

Everything else is proved: independence and the denoising tails (Lemmas 1, 2), soundness of decisive decisions (Lemma 3, assumption-free), the union-rate hazard the populations answer (Lemma 4), and certification soundness (Lemma 5, Propositions 1, 2) — the last self-contained up to the exact-test refinement of Remark 1, which cites Hoeffding (1956).